

Nutrition & Health: A Comprehensive Guide

This guide covers the fundamentals of nutrition science, dietary guidelines, the role of macronutrients and micronutrients, common dietary patterns, and evidence-based recommendations for maintaining optimal health.

1. Macronutrients

1.1 Carbohydrates

Carbohydrates are the body's primary energy source, providing 4 calories per gram. They are classified as simple (sugars) or complex (starches and fiber). The brain requires approximately 120 grams of glucose per day to function optimally.

Dietary fiber, a type of carbohydrate that cannot be digested, is essential for gut health. Soluble fiber (oats, beans) lowers cholesterol and blood sugar. Insoluble fiber (whole grains, vegetables) promotes digestive health. Adults need 25-38 grams of fiber daily.

1.2 Proteins

Proteins provide 4 calories per gram and are essential for building and repairing tissues, producing enzymes and hormones, and supporting immune function. Complete proteins (meat, fish, eggs, dairy, soy) contain all nine essential amino acids.

The Recommended Dietary Allowance (RDA) for protein is 0.8 grams per kilogram of body weight for sedentary adults. Athletes and older adults may need 1.2-2.0 grams per kilogram. Excessive protein intake above 3.5 g/kg/day may stress the kidneys.

1.3 Fats

Dietary fats provide 9 calories per gram and are essential for absorbing fat-soluble vitamins (A, D, E, K), hormone production, and brain health. Unsaturated fats (olive oil, avocados, nuts) are associated with reduced cardiovascular disease risk.

Trans fats, formed through partial hydrogenation of oils, increase LDL cholesterol and decrease HDL cholesterol. The WHO recommends eliminating industrial trans fats from the food supply. Many countries have banned or restricted trans fats.

1.4 Macronutrient Summary

Nutrient	Cal/gram	RDA	Key Functions	Best Sources
Carbohydrates	4	45-65% of calories	Energy, brain function	Whole grains, fruits, legumes

Protein	4	0.8g/kg body weight	Tissue repair, enzymes	Meat, fish, legumes, dairy
Total Fat	9	20-35% of calories	Vitamins, hormones, brain	Olive oil, nuts, avocado
Saturated Fat	9	<10% of calories	Cell membrane structure	Butter, coconut oil (limit)
Fiber	0	25-38g/day	Gut health, cholesterol	Vegetables, whole grains

2. Key Vitamins and Minerals

Nutrient	RDA (Adult)	Deficiency Risk	Food Sources
Vitamin D	600-800 IU	Bone loss, immune issues	Sunlight, fatty fish, fortified milk
Vitamin B12	2.4 mcg	Anemia, nerve damage	Meat, fish, dairy, eggs
Iron	8-18 mg	Anemia, fatigue	Red meat, legumes, spinach
Calcium	1000-1200 mg	Osteoporosis	Dairy, leafy greens, tofu
Omega-3	1.1-1.6 g	Heart disease risk	Fatty fish, flaxseeds, walnuts
Magnesium	310-420 mg	Muscle cramps, anxiety	Nuts, seeds, dark chocolate
Zinc	8-11 mg	Immune dysfunction	Meat, shellfish, legumes
Folate	400 mcg	Birth defects, anemia	Leafy greens, legumes, citrus

3. Popular Dietary Patterns

Mediterranean Diet: Emphasizes fruits, vegetables, whole grains, legumes, olive oil, and fish. Associated with reduced risk of heart disease, type 2 diabetes, and cognitive decline. Considered by many nutrition experts as the gold standard dietary pattern.

DASH Diet (Dietary Approaches to Stop Hypertension): Designed to lower blood pressure, emphasizing fruits, vegetables, whole grains, and low-fat dairy while limiting sodium, red meat, and added sugars. Consistently ranked among the healthiest diets.

Plant-Based Diet: Focuses on foods from plant sources. Can be vegetarian (no meat) or vegan (no animal products). Associated with lower rates of heart disease, cancer, and type 2 diabetes. Requires careful planning to ensure adequate B12, iron, and omega-3.

Ketogenic Diet: Very low carbohydrate (under 50g/day), high fat, moderate protein diet that induces ketosis. Effective for short-term weight loss and epilepsy management. Long-term safety and sustainability remain areas of ongoing research.

4. Evidence-Based Recommendations

Eat a variety of colorful vegetables and fruits daily — aim for 5-9 servings. Each color provides different phytonutrients and antioxidants that protect against chronic disease. Dark leafy greens are among the most nutrient-dense foods available.

Limit ultra-processed foods. Foods with long ingredient lists containing artificial additives, preservatives, and refined ingredients are associated with increased risk of obesity, cardiovascular disease, and certain cancers.

Stay hydrated. The general recommendation is about 3.7 liters (125 oz) for men and 2.7 liters (91 oz) for women of total water daily from all beverages and foods. Needs increase with physical activity, heat, and illness.

Limit added sugars to less than 10% of daily calories (WHO recommendation). Excess sugar consumption is linked to obesity, type 2 diabetes, dental caries, and non-alcoholic fatty liver disease.